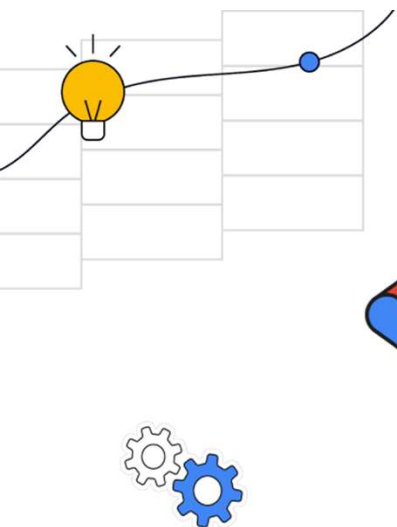




Solution Challenge

Powered by
H2S



Team Details

Team name: NEXUS THINKERS

Team leader name: S.V.S.S Vinod Kumar

Problem Statement: Many citizens face challenges in reporting and resolving civic issues such as garbage collection, road maintenance, water supply disruptions, and other public services. The existing communication channels with government authorities are inefficient, lack transparency, and result in slow response times. This leads to unresolved problems and dissatisfaction among residents.

Solution:

Civic Connect, an app that enables citizens to report civic issues in real-time, track the status of their complaints, and engage with local authorities through a transparent and efficient system. It incorporates features such as image-based reporting, geotagging, priority escalation, and community upvoting to ensure timely resolution and accountability.

One app for every solution.....

How is it different from existing ideas?

- Unlike traditional complaint systems, Civic-Connect offers real-time tracking, direct communication, and a transparent dashboard.
 - AI-based categorization for faster issue resolution.
- Community engagement through upvotes and feedback.

How will it solve the problem?

- Provides a structured and digitalized complaint registration system.
- Ensures faster responses by auto-routing complaints to the appropriate department.
- Keeps users informed with status updates and estimated resolution timelines.

Unique Selling Proposition (USP)

- Real-time issue tracking and notifications.
- AI-driven categorization and prioritization.
- Multi-language support for accessibility.
- Incentives and gamification to encourage civic participation.

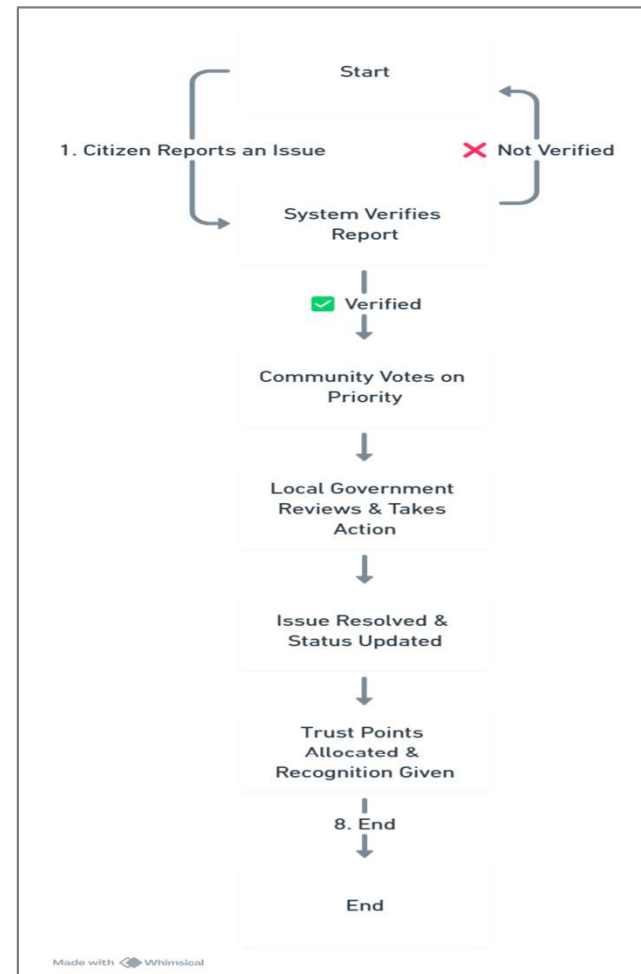
List of features:

- ✅ **One-Click Issue Reporting with Image Upload** – Quickly report issues with visual evidence.
- 🤖 **AI-Based Issue Categorization (Coming Soon!)** – Automated classification of complaints for faster resolution.
- 🔔 **Live Tracking & Real-Time Notifications** – Stay updated on the status of your complaints instantly.
- 🗣️ **Community Upvoting for Prioritization** – Let public votes decide which issues need urgent attention.
- ✉️ **SMS-Based Complaint Registration** – Report issues even without internet access!
- 🌐 **Multilingual Support (Coming Soon!)** – Ensuring inclusivity with multiple language options.
- 📊 **Transparency Dashboard** – Monitor complaint resolutions with complete visibility.
- 🏆 **Earn Trust Points** – Get rewarded for submitting valid complaints responsibly.
- 🏛️ **Access Government Services in One Place** – A single platform for all essential services.

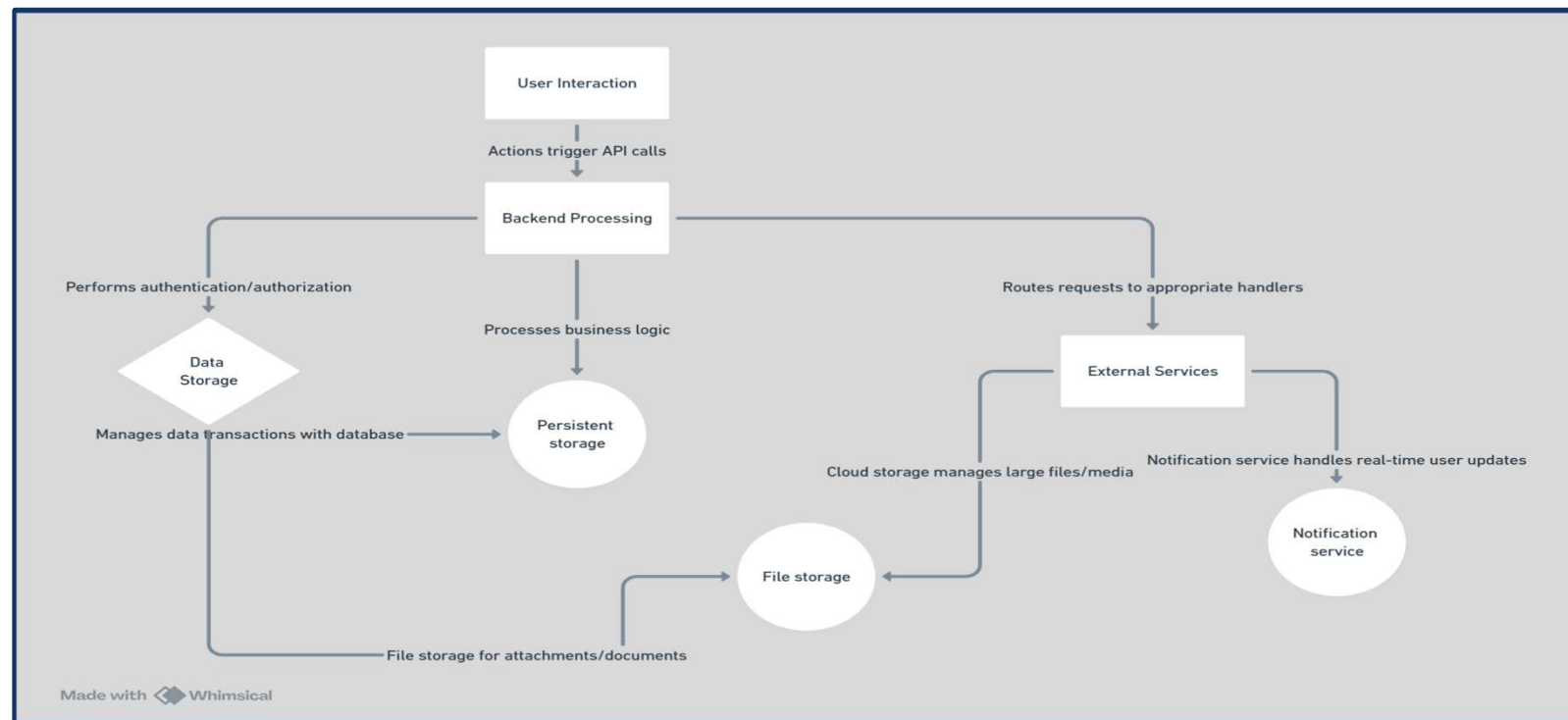
Process flow diagram or Use-case diagram

Technical Components Involved:

- **Authentication:** JWT tokens with refresh flow
- **APIs:** REST endpoints for CRUD operations
- **Real-Time:** WebSocket/Socket.io for chat and live tracking
- **Security:** HTTPS encryption, input sanitization
- **Notifications:** Twilio (SMS), SendGrid (Email)
- **File Handling:** Presigned URLs for S3 uploads



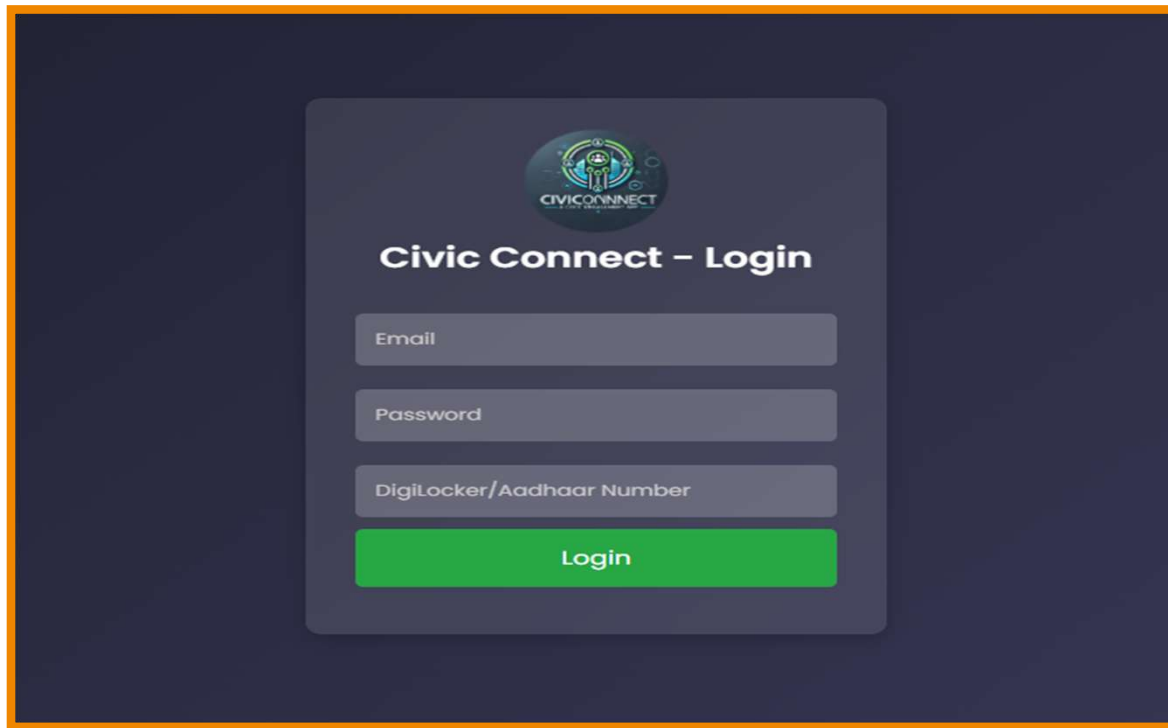
Architecture diagram of the proposed solution



Technologies to be used in the solution

- Frontend: React Native (for mobile), React.js (for web)
- Backend: Node.js with Express
- Database: Firebase / MongoDB
- APIs & Integrations: Google Maps API, AI-based image recognition (optional), Gemini APIs (for additional points)
- Cloud Deployment: Google Cloud / Firebase Hosting

Snapshots of the MVP



The image shows a login interface for 'Civic Connect'. It features a dark blue background with a central light blue rounded rectangle containing the login form. At the top of the form is the 'CIVICCONNECT' logo, which includes a stylized figure and the text 'CIVICCONNECT' and 'Your Smart Neighborhood'. Below the logo, the title 'Civic Connect - Login' is displayed. The form contains three input fields: 'Email', 'Password', and 'DigiLocker/Aadhaar Number'. A green 'Login' button is positioned at the bottom of the form.


CIVICCONNECT
Your Smart Neighborhood

Civic Connect - Login

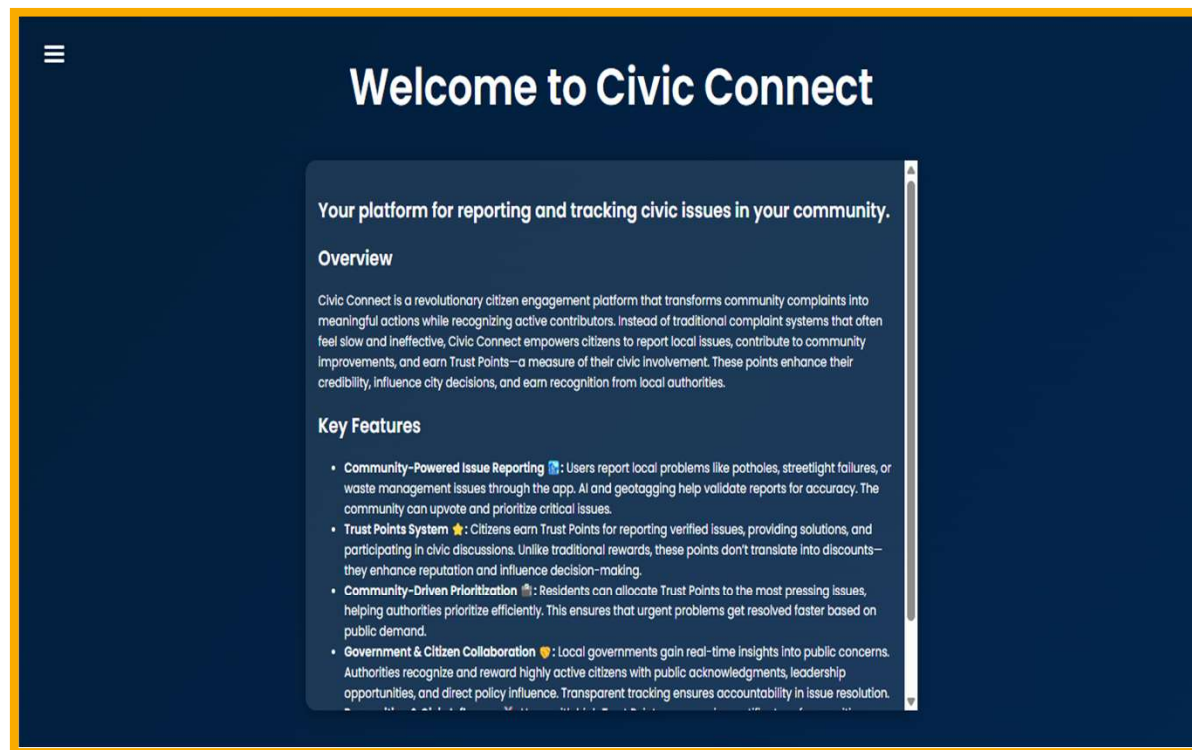
Email

Password

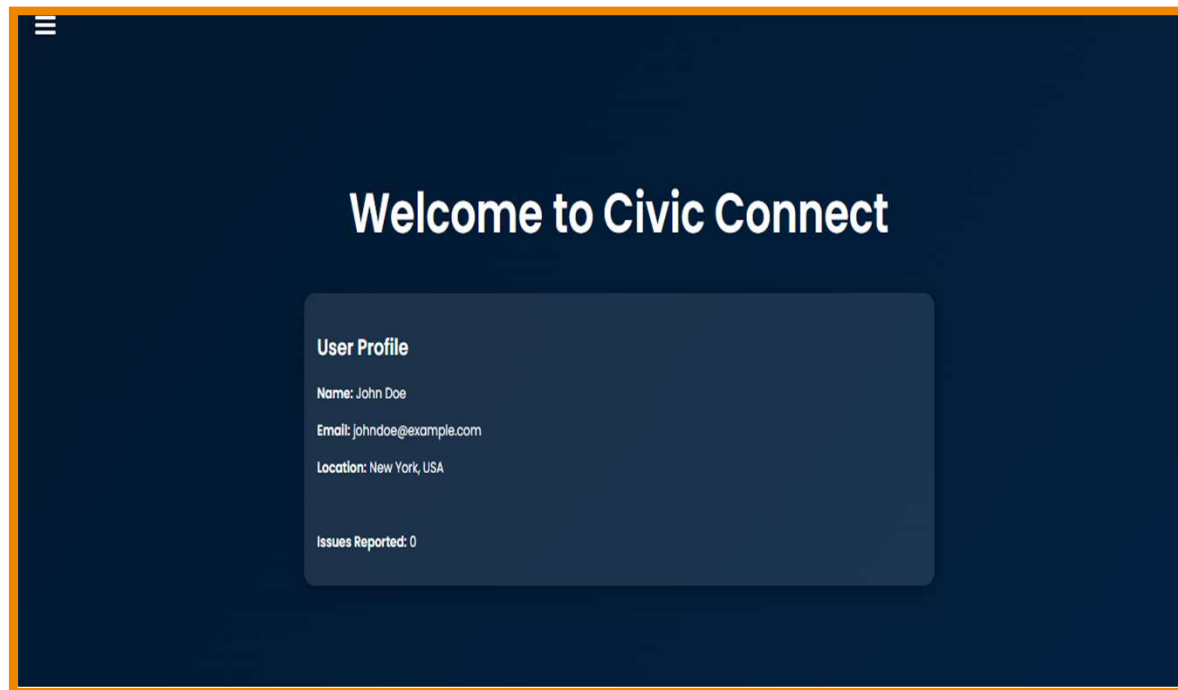
DigiLocker/Aadhaar Number

Login

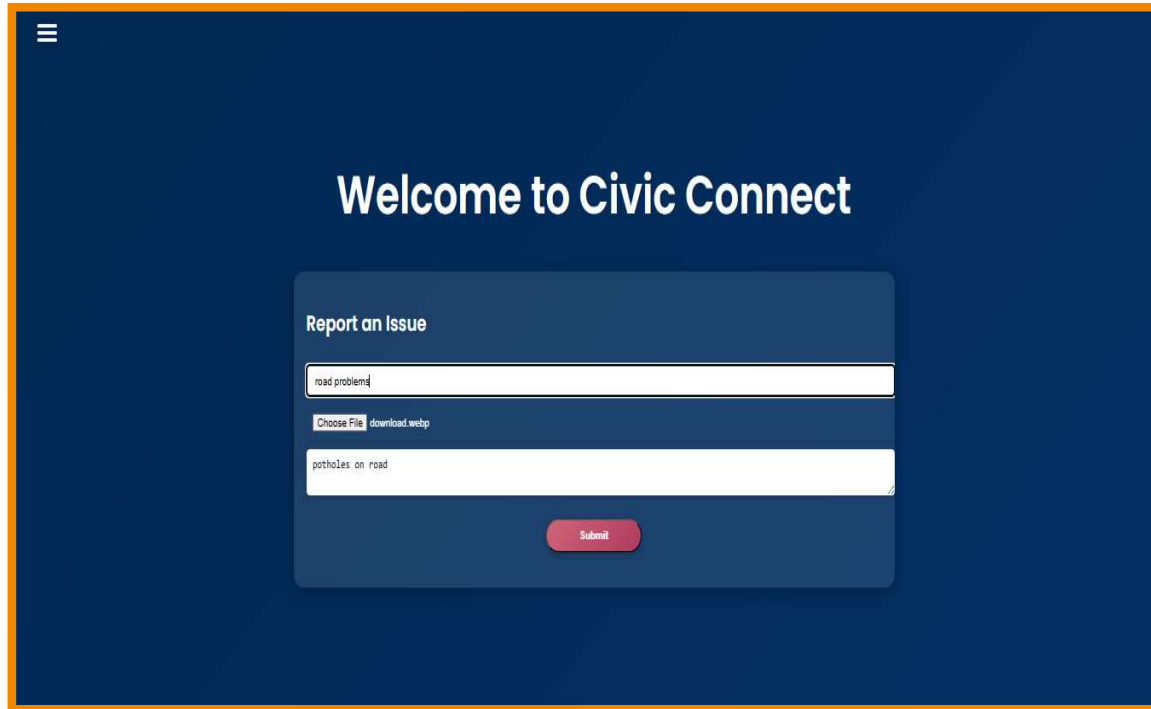
Homepage



Profile page



Issue Reporting Menu



The screenshot shows a web interface for reporting issues. It has a dark blue background with a white hamburger menu icon in the top left corner. The main heading is "Welcome to Civic Connect" in white. Below this is a "Report an Issue" section with a light blue background. This section contains two text input fields: the first has "road problem" and the second has "potholes on road". Between the fields is a file upload area with a "Choose File" button and a "download.webp" link. At the bottom of the form is a red "Submit" button.

☰

Welcome to Civic Connect

Report an Issue

road problem

Choose File download.webp

potholes on road

Submit

Community Voting



Welcome to Civic Connect

Current Issue List

Potholes -1

votes

Upvote

Downvote

Garbage Issues -1

votes

Upvote

Downvote

Water Supply Issues -0

votes

Upvote

Downvote

Electricity Problems -0

votes

Upvote

Downvote

Potholes

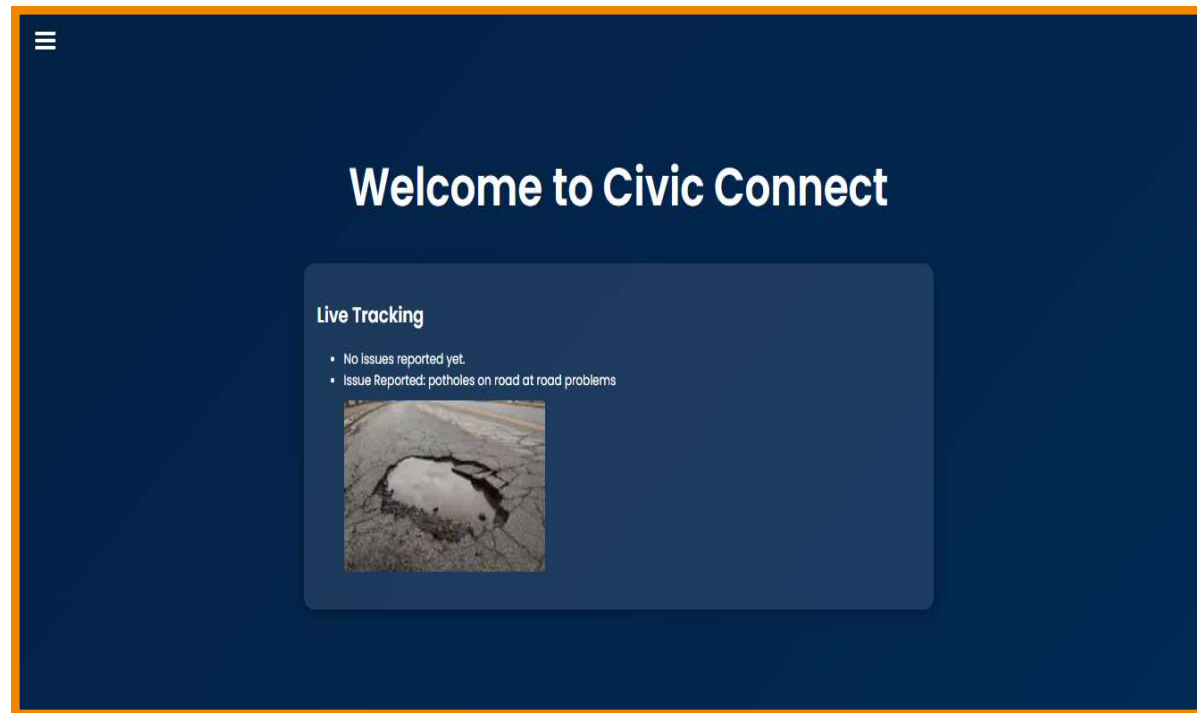
Garbage

Water Supply

Electricity



Live tracking of Reported Issues



GitHub Public Repository :

https://github.com/Vinod63021/Civic_Connect/actions

Demo Video Link (3 Minutes) :

https://drive.google.com/file/d/16_e9tSac6VNbyyGFl-OgvA48kjPaTsmM/view?usp=sharing

MVP Link :

<https://drive.google.com/file/d/1PTad1mXbWHOPrt40uVSKuz7n5sdg-qQu/view?usp=sharing>



Solution Challenge



Thank you

